Malvika Raj Joshi

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Research Interests: Quantum computing, quantum-classical separations, privacy & cryptography.

Education

University of California, Berkeley 2021–Present
Ph.D. in Computer Science. Advisors: Umesh Vazirani & John Wright
Research area: Quantum complexity theory.

Massachusetts Institute of Technology 2020–2021
M.Eng. in EECS. Advisor: Aram Harrow.
Thesis: *Pretending to be Quantum: A Study of IQP-Based Tests of Quantumness*

Massachusetts Institute of Technology 2016–2020
B.S. in Physics and B.S. in Computer Science. GPA: 5.0/5.0.

Publications

- Lucas Gretta, Meghal Gupta, and **Malvika Raj Joshi**. *Parity* $\notin \mathsf{QAC}^0 \iff \mathsf{QAC}^0$ *is Fourier-Concentrated.* (FOCS, 2026).
- Lucas Gretta and **Malvika Raj Joshi**. *Shor's Algorithm Requires Fanout*, 2026. arXiv:2608.06703.
- Lucas Gretta, Meghal Gupta, and **Malvika Raj Joshi**. *Polylogarithmic-Weight Dicke States in* QAC^0 *and Arbitrary Symmetric States in* QAC^0_f, 2026. arXiv:2604.15298.
- **Malvika Raj Joshi**, Avishay Tal, Francisca Vasconcelos, and John Wright. *Improved Lower Bounds for* QAC^0. (STOC, 2026).
- **Malvika Raj Joshi** and Francisca Vasconcelos. *Constant-Depth Unitary Preparation of Dicke States*, 2026. arXiv:2601.10693.
- Shyan Akmal, Lijie Chen, Ce Jin, **Malvika Raj**, and Ryan Williams. *Improved Merlin–Arthur Protocols for Central Problems in Fine-Grained Complexity.* (ITCS, 2022).

Industry Experience

Amazon, East Palo Alto — *Applied Scientist Intern* May–Aug. 2026
Differentially-Private Diffusion with improved utility. Preprint in preparation.

Amazon, East Palo Alto — *Applied Scientist Intern* May–Aug. 2025
Designed *Lexicure*, a leakage-free encrypted lexical search protocol, and developed new secure primitives for higher-degree arithmetic. *With:* Tal Wagner, Shai Halevi, Nina Mishra. Patent pending.

Facebook, New York, NY — *Software Engineering Intern (PyTorch)* May–Aug. 2019
Designed and implemented a C++ API for PyTorch Autograd introducing native C++ support for automatic differentiation.

Facebook, Menlo Park, CA — *Software Engineering Intern* May–Aug. 2018
Optimized the News Feed ranking pipeline by parallelizing components in C++.

WhatsApp, Menlo Park, CA — *Software Engineering Intern* Jun.–Sep. 2017
Designed and implemented a caching mechanism for encrypted downloads in backend infrastructure.

Honors and Awards

- **International Olympiad in Informatics (IOI)**: Silver (2014, 2015; ranks 28, 29); Bronze (2016).
- **Asia Pacific Informatics Olympiad (APIO)**: Silver (2014); Bronze (2015, 2016).
- Elected member of **Sigma Pi Sigma** (Physics Honor Society).
- Microsoft Q# Coding Contest 2019: Top 50.

Outreach and Professional Activities

INOI Training Initiative for Women 2020–2021

Founder and Volunteer

Organized and taught algorithms and programming to female high school students preparing for the Indian National Olympiad in Informatics (INOI). Developed curriculum, delivered lectures, and provided mentorship.

Massachusetts Institute of Technology June–Aug. 2020

Research Intern, CSAIL (Advisor: Dina Katabi)

Processed large-scale video and RF data for sleep monitoring, designed and implemented caching mechanism for Gunicorn-based server.

Competitive Programming 2015-2017

Problem Setter

Problem setter for programming contests including Indian IOI selection rounds (IOITC, ZIO/ZCO), ACM ICPC Chennai Regionals, and CodeChef Long Challenges.

Massachusetts Institute of Technology Oct.–Dec. 2016

UROP Researcher (Advisor: Nancy Kanwisher)

Developed components of an iOS application for human eye-tracking experiment.

Teaching Experience

University of California, Berkeley

Graduate Student Instructor (CS70 Fall 2021, CS170 Fall 2023, CS270 Fall 2025)

Led discussions, office hours, and graded coursework; additionally created homeworks and exams (CS170).

Massachusetts Institute of Technology

Grader, 6.045: Automata, Computability, and Complexity Theory (Sp. 2018, 2020)

Graded coursework and exams.

Skills

Codeforces International Master (peak 2318); **CodeChef** 7-star (peak 2552); **Topcoder** top 4%.

Programming languages: C, C++, Python, Java, Assembly.